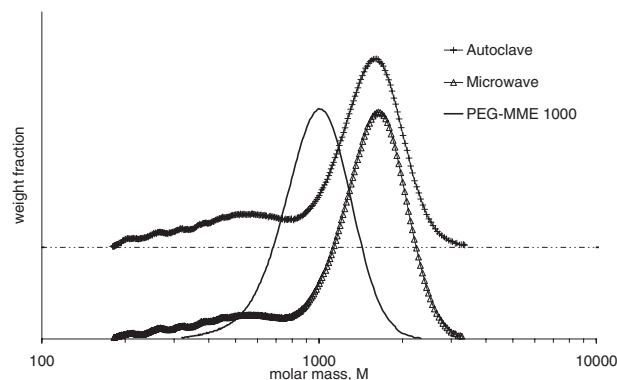


Selectivity of PEO-*block*-PPO Diblock Copolymers in the Microwave-Accelerated, Anionic Ring-Opening Polymerization of Propylene Oxide with PEG as Initiator

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Diblock copolymers of poly(ethylene oxide) (PEO) and poly(propylene oxide) (PPO) were synthesized by the anionic ring-opening polymerization of propylene oxide (PO), with controlled microwave heating in sealed vessels. A detailed study was carried out to investigate the effects of different parameters on the formation of unwanted byproducts. Parameters that were considered include temperature; the concentration of NaH, monomer and hydroxy groups in the feed; and the polarity of the reaction medium. A continuous decrease of internal pressure during the sealed-vessel experiment reflected the consumption of PO monomer and the completion of the reaction was confirmed by a drop of the internal pressure to zero when reactions were performed in bulk. The products were characterized by using different chromatographic techniques. A comparison of the reaction times and composition of the polymers prepared by microwave and conductive heating is given.



Introduction

Amphiphilic block copolymers consist of hydrophilic and hydrophobic blocks, which are linked to each other by covalent bonds. Consequently, they exhibit surface activity. Recently these materials have attracted considerable attention due to their applications in several different

fields: for the modification of surfaces to control protein adsorption,^[1] in nano- and biomaterials,^[1] for controlled release,^[2] for nanofiltration membranes, for the surface modification of biomaterials,^[3] as synthetic lubricants, as emulsifiers, and for biomedical applications.^[1,4]

In most cases, poly(ethylene oxide) (PEO) is the hydrophilic block, while alkyl, aryl alkyl, polyether, polyester, or polystyrene, for example forms the hydrophobic block. Additionally, there has been much work on triblock copolymers of PEO and poly(propylene oxide) (PPO) with PPO as the central block (commercially known as poloxamers).^[5,6] Some studies on the synthesis and applications of poly(ethylene oxide)-*block*-poly(propylene oxide) (PEO-*block*-PPO) diblock copolymers has also been reported.^[7–9]

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Microwave dielectric heating is a very popular technology in organic synthesis, and to some extent also in polymerization reactions.^[10] It has many advantages compared to conventional heating applying conduction phenomena, such as the rapid and uniform heating. In some instances it has been shown to reduce reaction times from hours to minutes (or even seconds), or from days to hours. Schubert et. al.^[11,12] published comprehensive reviews about microwave-assisted polymerizations. These reviews cover almost all classes of polymerizations and refer to ring-opening polymerizations of ϵ -caprolactone,^[13] ϵ -caprolactam and 2-ethyl-2-oxazoline.^[14] Recently, Schubert and coworkers reported the microwave-assisted synthesis of block copolymers of poly(2-oxazoline).^[15,16]

The microwave-assisted generation of triblock copolymers of poly(ethylene oxide) with polylactide^[17] and poly(ϵ -caprolactone)^[18] has also been reported. To the best of our knowledge the production of PEO-*block*-PPO block copolymers under microwave irradiation conditions has not yet been reported.

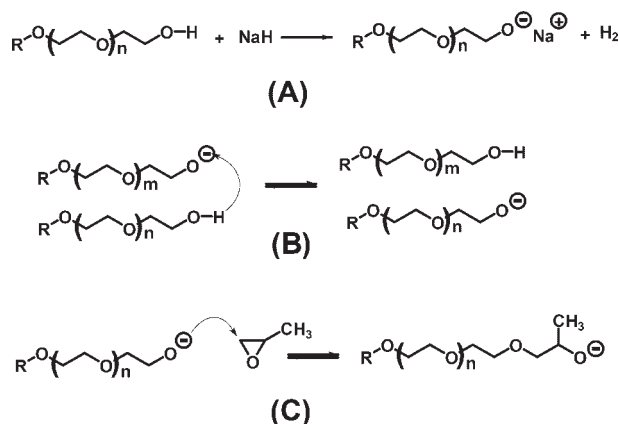
In this context, we have tried to synthesize diblock copolymers of PEO-*block*-PPO by the anionic ring-opening polymerization of propylene oxide (PO) using controlled microwave dielectric heating.

As has already been described by several authors,^[19–22] there is a significant side reaction in the anionic polymerization of propylene oxide, which leads to the formation of polyoxypropylene chains with one terminal allyl group. This chain-transfer reaction produces an amount of homopolymer instead of the desired block copolymers. In the above-mentioned publications, the influence of the reaction conditions has been studied, but in most of these studies only the overall content of the unsaturated groups was determined: no information on the molar-mass distribution of the different homologous series was given. We have now performed a detailed study on the different parameters affecting the quantity and molar mass of the unwanted homopolymers (both poly(propylene glycol) PPG diols and allylic PPOs).

Reaction Pathways and Side Products

In the polymerization of PO several elementary reactions have to be taken into account, as is shown in the following schemes.

Sodium hydride (in catalytic amounts) reacts with poly(ethylene glycol) PEG to yield the active chain-ends (reaction A in Scheme 1). As the exchange of protons between the active chain-ends and the terminal –OH groups of the PEG diol or PEG monomethyl ether (B) is very fast, all chains have the same probability to grow. Hence, the addition of PO proceeds at the ends of all PEG chains with the same rate (C).



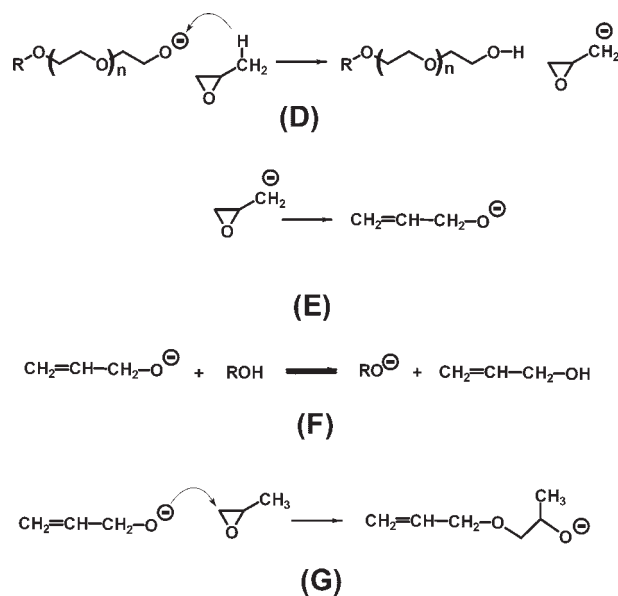
Scheme 1. Reaction A: Sodium hydride reacts with PEG to yield active chain-ends. Reaction B: Exchange of protons between the active chain-ends and the terminal –OH groups of the PEG diol or PEG monomethyl ether. Reaction C: The addition of PO proceeds at the ends of all the PEG chains

The active chain-ends may, however, react with PO in a chain-transfer reaction (reaction D in Scheme 2). The transfer of a proton terminates the growing chain (which can, however be activated again) and forms an unstable anion of PO, which rearranges to the anion of allyl alcohol (reaction E). Obviously, this reaction corresponds to an isomerization of propylene oxide to allyl alcohol, which creates a new hydroxy group, as the allyloxy anion can be reprotonated (F). The allyloxy anion starts a new chain by reacting with the monomer (G), thus yielding a homopolymer of PO, with a terminal allyl group, instead of the desired block copolymer.^[20,21] The degree of unsaturation can be determined by titration or by NMR spectroscopy.^[21,23]

As the chain transfer to the monomer occurs as long as there are living chains and residual monomer, chains with allylic end groups will be started all the time. Consequently, one must expect that low-molecular-weight oligomers are still formed at rather high conversion. This makes it rather easy to identify them as the low molecular fraction in size-exclusion chromatography (SEC). Homopolymers can, however, also be formed via a different pathway: in the presence of water, hydroxyl ions are formed (reaction H in Scheme 3), which react with the PO, yielding the (primary or secondary) anion of propylene glycol (PG).

As propylene glycol contains two hydroxy groups, it may be activated on both sides (I) and thus grow in both directions. Consequently, the poly(propylene glycol) (PPG) will grow faster than the allylic homopolymer and the desired diblock copolymer.

If the reaction mixture already contains moisture at the beginning (from insufficiently-dried reagents or from the surface of the glassware), the PPG thus formed may



Scheme 2. Reaction D: The active chain-ends may react with PO in a chain-transfer reaction. Reaction E: The transfer of a proton terminates the growing chain (and forms an unstable anion of PO, which rearranges to the anion of allyl alcohol). Reaction F: The allyloxy anion can be reprotated. Reaction G: The allyloxy anion starts a new chain by reacting with the monomer.

have a rather high molar mass, in some cases as high or even higher than the block copolymer. Consequently, it cannot be identified by size-exclusion chromatography, thus it is highly important to combine different chromatographic techniques in the characterization of such products. In particular, this is a combination of dual detector SEC and liquid chromatography at critical conditions (LCCC).

As has been shown by Becker,^[20] the rates of chain propagation R_p (which corresponds to that of the forma-

tion of O–C–O bonds) and chain transfer R_c to monomer (the formation of C=C bonds) depend on the concentrations of active centers (RO^-) and propylene oxide (PO), but with different rate constants k_p and k_c .

Consequently, the ratio of both reactions should be given just by the ratio of the rate constants, as the concentration of active centers and monomer cancel out:

$$\frac{R_c}{R_p} = \frac{k_c}{k_p} = C \quad (1)$$

Expressing the polymerisation of PO by Equation (1) is, however, only a first approximation.^[20]

In reality, this simple relation does not hold: Gladkovskij^[24] has proposed a different mechanism, in which also the concentration of the –OH groups has to be taken into account:

$$\frac{R_c}{R_p} = \frac{C}{[ROH]} \quad (2)$$

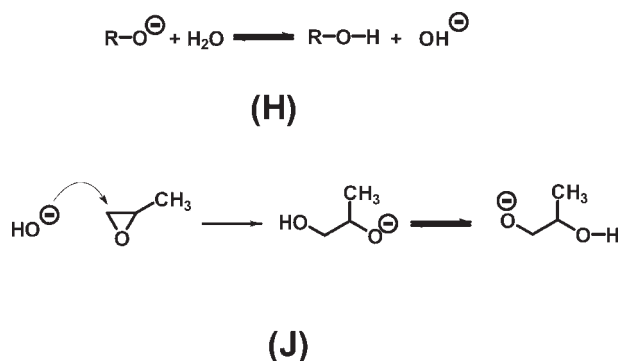
The influence of the –OH concentration is due to the association of the –OH groups with PO, by which the chain transfer becomes less likely than the addition of PO to the active chain-end. At high concentrations of –OH groups (or low monomer concentration) the formation of allyl groups should be suppressed, while at low –OH concentrations Equation (2) describes the reaction quite well.^[20]

In this study we have tried to apply these findings to the rapid synthesis of PEO-*block*-PPO block copolymers under controlled microwave-irradiation conditions and compare the results with those obtained with conductive heating. For this purpose we had to develop efficient and reliable techniques for the characterization of the products.

Characterization of the Products

Functional polymers and oligomers can be characterized using different chromatographic techniques, which separate according to different criteria. The type of chromatographic separation is determined by the so-called interaction parameter c , which describes the interaction of a structural unit with the stationary phase.^[25]

This parameter is negative in size exclusion chromatography (SEC) and positive in liquid adsorption chromatography (LAC). At the critical adsorption point (CAP) the interaction parameter equals zero, and the corresponding block becomes “chromatographically invisible”.^[26–29] This effect is utilized in liquid chromatography under critical conditions (LCCC), which allows the separation of a polymer homologous series according to their end-groups.^[27,30–32] In the case of amphiphilic polymers the interaction parameters c_A and c_B of the individual structural units (A and B)



Scheme 3. Reaction H: Formation of hydroxyl ions in the presence of water. Reaction J: Hydroxyl ions may react with PO, yielding an anion of propylene glycol. The propylene glycol may be activated on both sides, given its two hydroxy groups, and thus grow in both directions.

may have different values. If both c_A and c_B are negative, the molar-mass distribution (MMD) can be determined by size exclusion chromatography (SEC), with retention decreasing with increasing molar mass, and only a minor influence of chemical composition. SEC yields the overall MMD; with dual detection^[33–35] one may obtain information on the chemical composition along the MMD.^[35–37] SEC is, however, not capable of discriminating a mixture of block copolymers and homopolymers.

At the CAP for the ethylene oxide (EO) unit, a separation according to the PO block(s) may be achieved by an SEC or LAC mechanism, while at the CAP for PO, the length of the EO block(s) may be obtained. The application of these methods to PEO-*block*-PPO block copolymers has been described recently.^[38,39] In the case of PPG one may expect, that, at the CAP for PO, a separation of the (non-functional) PPG diols and the allyl-terminated polyoxypropylenes may be possible if the interaction of the allyl group with the stationary phase is sufficiently strong.

If the peak of the PPG diols and the solvent peak overlap (as can be expected in LCCC of non-functional chains), quantification is impossible with density or refractive-index (RI) detection. In this case, an evaporative light-scattering detector (ELSD) can be used, which does not see the solvent peak. Unfortunately, it also does not detect the lowest-molecular-weight oligomers of many polymers,^[40–42] which may be present in the sample, especially for PPG with allylic end-groups.

Moreover, one must expect a molar-mass dependence of the response factor of any detector in the range of lower-molecular-weight oligomers (due to the increasing influence of the end-groups).

The problem of quantification in LCCC shall be the subject of another communication.

Experimental Part

Polymerizations

Propylene oxide (PO) and poly(ethylene glycol) monomethyl ethers (PEG-MME) with different molecular weights (550, 1 100, 2 000 g · mol⁻¹) were purchased from Fluka (Buchs, Switzerland). A few experiments were performed with a new batch, which was specified as PEG-MME 1 000.

Solvents (toluene, dioxane, benzene, and cyclohexane) were purchased from Merck (Darmstadt, Germany). Sodium hydride (NaH) (60% dispersion in mineral oil) from Aldrich. PEG-MME was dried by azeotropic distillation with toluene. PO was refluxed over calcium hydride (Fluka).

The polymerization of PO was performed by anionic ring opening using PEG-MME and sodium hydride as initiator. All polymerizations were performed in 2–5 ml (filling volume) microwave process vials (Biotage AB), which were oven-dried under vacuum prior to use. PEG-MME and NaH were added to the vials which were subsequently sealed with an aluminum

crimp-top and a teflon septum. After purging the vials with argon for 5 minutes, the calculated volume of PO and solvent were added with a syringe through the septum. The sealed microwave process vials were subsequently introduced into the cavity of the single-mode microwave reactor (Biotage Initiator Eight EXP 2.0, absorbance level: high).

All of the microwave-assisted polymerizations were carried out at temperatures between 130 and 175 °C to complete conversion, as indicated by a drop of the internal pressure from initially 10–20 bar to zero in bulk, which was reached after 20 to 180 min depending on the temperature, catalyst concentration and the length of the PPO block.

In the case of solution polymerization there was no significant drop of pressure because of the vapour pressure of the solvent. For comparison, all of the solution polymerizations were carried out at 160 °C for three hours.

After polymerization the active end was neutralized by adding an equimolar quantity of acetic acid. In Table 1 and Table 2 the protocols for all of the syntheses are summarized in bulk and in solution respectively. In the case of the solution polymerizations the solvent was removed after the polymerization on a rotary evaporator.

For comparison, polymerization with conductive heating was performed in an 250 ml autoclave H16296 from Roth (Karlsruhe, Germany) equipped with an internal temperature probe (the limits of temperature and pressure were 300 °C and 200 bar respectively). Teflon seals and stirrers were used. The reaction was performed with same ratios of reactants and conditions as in experiment #8 in Table 1.

The desired temperature was reached after 20 min; the initial pressure was 18 bar. The reaction proceeded more slowly than with microwave heating: it took almost 5 hours to completion (monitored by decrease of pressure to zero) compared to 70 min in microwave heating.

As a reference material, allyl-terminated PPG was synthesized by microwave-assisted polymerization of PO at 160 °C using allyl alcohol with sodium hydride as the initiator. The polymerization procedure was the same. The only difference was that allyl alcohol was used instead of PEG-MME. All of the other protocols for the reaction are given in Table 3.

Chromatography

These investigations were performed using a DDS70 density detector (CHROMTECH, Graz, Austria) (according to the mechanical oscillator principle). Data acquisition and processing was performed using the CHROMA software, which has been developed for the DDS70.

In SEC, we used a modular system consisting of a Gynkotek 300C pump, a VICI injector equipped with a 100 µl sample loop, two Rheodyne 7060 column selection valves (Rheodyne, Cotati, CA, USA), a DDS 70 density-detection system (Chromtech, Graz, Austria) coupled with a Bischoff 8110 refractive-index detector (Bischoff, Leonberg, Germany).

SEC measurements were performed on a 7.8 × 600 mm Phenogel 100A column (Phenomenex, Torrance, CA, USA) or a 7.8 × 300 mm PLgel MIXED E column (Polymer Laboratories, Church Stretton, Shropshire, UK) at a flow-rate of 1.00 ml · min⁻¹. Sample concentrations were between 3.0 and 10.0 g · l⁻¹. The

Table 1. Protocols for the microwave-assisted synthesis of PEO-*block*-PPO diblock copolymers in bulk.

Experiment number ^{a)}	Mass of PEG-MME	Mass of PO	Mass of NaH	PO/PEG-MME mass ratio	Temperature	Initial pressure	Final pressure	Time
	g	g	mg		°C	bar	bar	min
1	1.5	1.5	7.5	1.00	160	17	2	20
2	1.5	1.5	5.0	1.00	140	11	0	120
3	1.5	1.5	5.0	1.00	175	19	0	30
4	1.5	1.5	5.0	1.00	130	9	1	180
5	1.5	1.5	5.0	1.00	160	16	0	55
6	1.5	1.5	2.0	1.00	160	16	0	140
7	2.1	0.9	3.0	0.43	160	12	0	50
8	1.5	1.5	3.0	1.00	160	15	0	75
9	1.5	1.5	3.0	1.00	160	16	1	55
10	1.5	1.5	3.0	1.00	160	14	2	90
11	0.9	2.1	3.0	2.33	160	18	1	120
12	1.8	1.2	3.0	0.67	160	14	4	75
13	1.2	1.8	3.0	1.50	160	16	1	130

^{a)}Experiment #9 was with PEG-MME 550, experiment #10 was with PEG-MME 2000.

columns were calibrated with PEG standards from Sigma-Aldrich and FLUKA (Buchs, Switzerland). Molar-mass distributions were calculated using CHROMA.

In LCCC, the mobile phase was delivered by an ISCO 2350 HPLC pump (ISCO, Lincoln, Ne, USA) at a flow rate of $0.5 \text{ ml} \cdot \text{min}^{-1}$. Samples were injected using a Spark SPH 125 Fix autosampler (Spark Holland, Emmen, The Netherlands) equipped with a $20 \mu\text{l}$ loop. An RI 2414 refractive-index detector (Waters) or a SEDEX 45 ELSD (Sedere, France) was connected to the DDS 70. Nitrogen was used as the carrier gas, and the pressure at the nebulizer was set to 1.0 bar. The evaporator temperature was 30°C .

In both systems, the columns and density cells were placed in a thermostatted box, in which a temperature of 25.0°C was maintained for all measurements in LCCC and 30.0°C in SEC.

Mobile phases were mixed by mass and vacuum-degassed, their composition was controlled by density measurements using a DMA 60 density meter equipped with a DMA 602 M measuring cell (A. Paar, Graz, Austria).

The following columns were used in LCCC:

Jordi Gel DVB 500 RP: 100% poly(divinylbenzene); $250 \times 4.6 \text{ mm}$; particle diameter = $5 \mu\text{m}$; nominal pore size = $500 \text{ \AA}$, (Jordi, Bellingham, MA, USA).

Discovery HS PEG, silica-based PEG phase; $250 \times 4.6 \text{ mm}$; particle diameter = $5 \mu\text{m}$; nominal pore size = $120 \text{ \AA}$ (Supelco, Bellefonte, PA, USA).

The solvents (chloroform, tetrahydrofuran (THF), acetone and water, all HPLC grade) were purchased from Roth (Karlsruhe, Germany) and Merck (Darmstadt, Germany). PEG and PPG

Table 2. Synthesis protocols in solution.

Experiment number	Mass of PEG-MME	Mass of PO	Mass of NaH	PO/PEG-MME mass ratio	Temperature	Solvent	Solvent concentration	Time
	g	g	mg		°C		%	min
14	1.0	1.0	2.0	1.00	160	dioxane	50	180
15	1.0	1.0	2.0	1.00	160	benzen	50	180
16	1.0	1.0	2.0	1.00	160	cyclohexane	50	180
17	1.0	1.0	2.0	1.00	160	toluene	50	180
18	1.0	1.0	2.0	1.00	160	toluene	33	180
19	1.0	1.0	2.0	1.00	160	toluene	60	180

Table 3. Protocols of allylic PPO synthesis and results (temp = 160 °C, amount of NaH = 3 mg).

Experiment number	Mass of allyl alcohol	Mass of PO	Average number of PO units	SEC results		
	g	g		$\overline{M}_w$	$\overline{M}_n$	$\overline{M}_w/\overline{M}_n$
20	0.5	2.00	4	294	263	1.1
21	0.3	2.25	8	548	461	1.2
22	0.2	2.30	11	723	601	1.2

standards were purchased from Sigma-Aldrich and FLUKA (Buchs, Switzerland).

Results and Discussion

In microwave-assisted batch-polymerization reactions using sealed vessels, generally all of the reagents have to be added at the beginning, since it is not easily possible to introduce reagents during the microwave experiment.^[10] This means that it is not possible to keep the monomer concentration low by adding the monomer continuously, which would largely suppress the formation of allylic chains.^[20,21] On the other hand, microwave-assisted polymerizations can be carried out conveniently in several minutes at comparatively high temperatures, compared to several hours using conventional processing.

A typical microwave experiment (Table 1, entry 8) was additionally performed under similar conditions, in a 250 mL steel autoclave equipped with an internal temperature probe employing conventional heating. Using conventional heating it took 5 hours for complete conversion while the same experiment under microwave irradiation took only 70 minutes. However, since the volume of the two reactors was vastly different (10 mL versus 250 mL) the results cannot be directly compared. While in the microwave experiment the designated, set reaction temperature of 160 °C was reached in less than 2 minutes; it took

approximately 45 min to ramp the internal reaction to 160 °C in the autoclave. Similar differences were also observed during the cooling process (Figure 1). We therefore believe the observed differences in product composition displayed in Figure 2 are a consequence of the different heating/cooling profiles and not the result of any nonthermal microwave effects.

At reaction temperatures between 130 °C and 175 °C the initial pressure in the microwave process vial was typically in a range between 10–20 bar and decreased during the polymerization as PO was consumed. Figure 1 shows a representative temperature-pressure-time graph of the progress of the reaction. In the insert (on a different time scale) the power consumption is also shown, which goes to a maximum in the first stage of the reaction, and

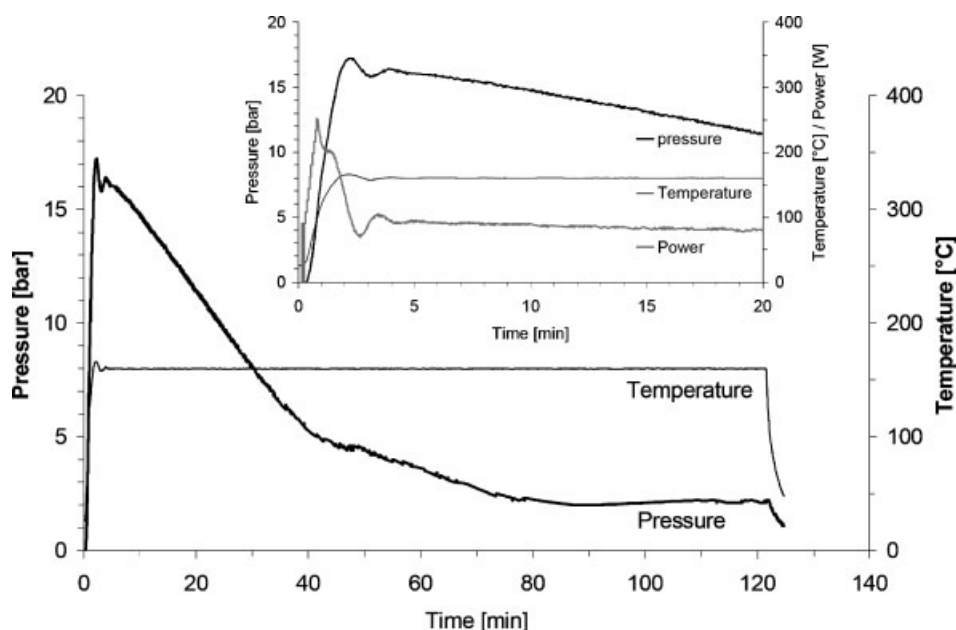


Figure 1. Temperature and pressure profile for the sealed-vessel microwave-assisted synthesis of PEO-*block*-PPO diblock copolymers (experiment #8, Table 1) with Microwave flash heating (0–250 W, 0–2 min), temperature control using the feedback from IR thermography (constant 160 °C, 2–120 min), and active cooling by compressed air (120–125 min). The pressure drop from around 16 to 2 bar indicates the progress of the reaction. Insert: Detailed view showing the temperature-pressure-power profile for the initial 20 min of microwave irradiation.

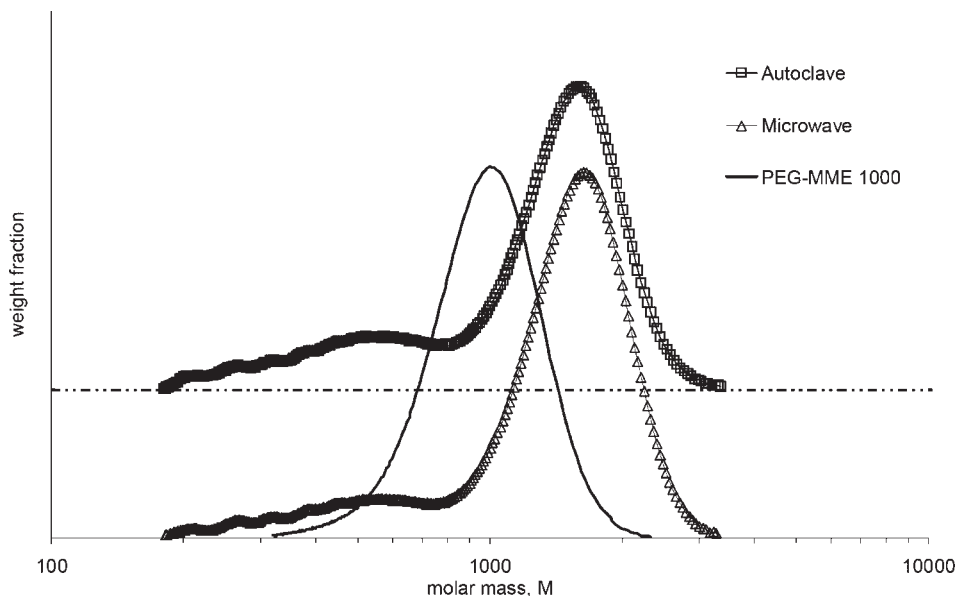


Figure 2. Effect of toluene concentration on MMD, as obtained from SEC on Phenogel 100 Å. Reaction conditions: experiments #17, 18, 19 in Table 2 and #8 in Table 1.

approaches a more or less constant value after several minutes.

Due to the fact that PO has a boiling point of only 34.3 °C the autogenic pressure generated by this starting material in combination with the online pressure monitoring of the microwave reactor^[10] did provide a very convenient method for monitoring the conversion in these solvent-free polymerization processes. A pressure drop to (nearly) zero would typically indicate a complete conversion, as confirmed by control experiments stopping the reaction at pressure levels of 0 to 3 bar. It should be mentioned, that a readout of 3 bar was observed sometimes even after cooling. The reason for this is that the pressure is measured through the septum on top of the vial, which sometimes maintains some deformation, even when the pressure in the vial is already zero.

The products thus obtained were characterized using different chromatographic techniques: SEC with dual detection yields information on chemical composition along the molar-mass distributions. Of course SEC is not able to differentiate between a copolymer and a mixture of

homopolymers with the same hydrodynamic volume, but in our case the homopolymers with allylic end-groups should be considerably smaller than the block copolymer, as the starting block (PEG-MME) is already quite large.

This is indeed the case, as can be seen from Figure 3, which shows a comparison of the chromatograms (from SEC with density detection) of the starting block and a copolymer sample containing a considerable amount of byproducts. The main fraction of the copolymer sample obviously elutes earlier (at 12.0 ml) than the starting block (PEG-MME 1 100, at 12.5 ml), but there is also a fraction with a lower molar mass than the starting material, which elutes at 14.0 ml (The peak at 18.8 ml is the peak of the internal standard (THF), which was used for flow-rate correction.)

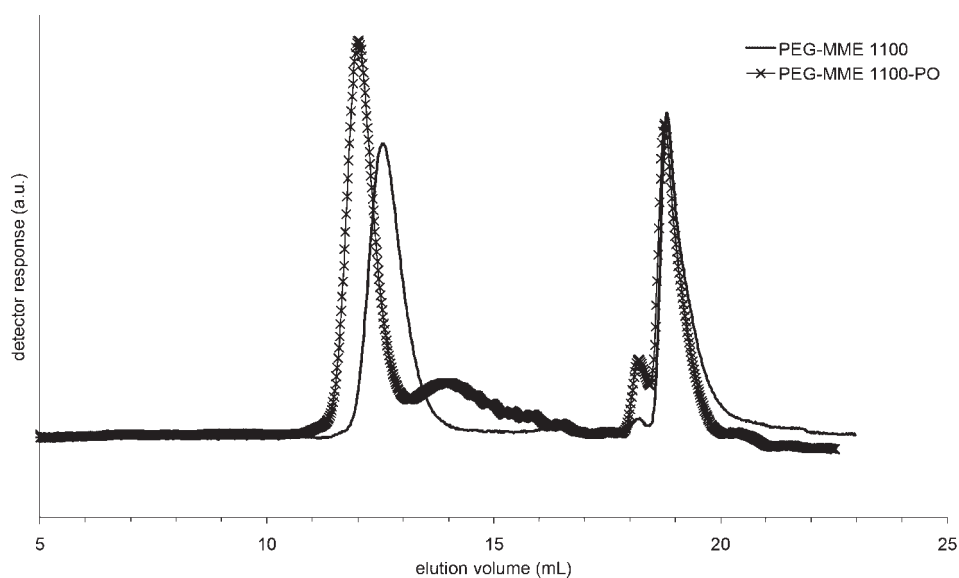


Figure 3. Overlay of SEC chromatograms of PEG-MME 1 100 and a copolymer sample: Phenogel 100A, 1.0 ml · min⁻¹, density detection.

Figure 4 shows the MMD and the chemical composition of another sample: the lower molecular fraction obviously consists of PPG homopolymer.

The MMD of the sample can be divided into copolymer and homopolymer parts (confirmed by the chemical composition of each fraction from dual detector SEC). If the starting block is too small (PEG-MME 550), this is not possible, as the fractions overlap. With PEG-MME 1 100 the fractions are sufficiently separated, and with and PEG-MME 2 000 the fractions do not overlap any more (Figure 5). In this case the molar mass of the copolymer is too high for the 100 Å column; hence we used a PLgel Mixed E column for SEC of this product.

As the 100 Å column has a much better resolution in the lower molecular-weight range, we have chosen PEG-MME 1 100 as the starting block for most of the experiments in this study.

From dual-detector SEC the absolute amount and the chemical composition of both fractions can be determined, and therefrom the percentage of the converted PO,

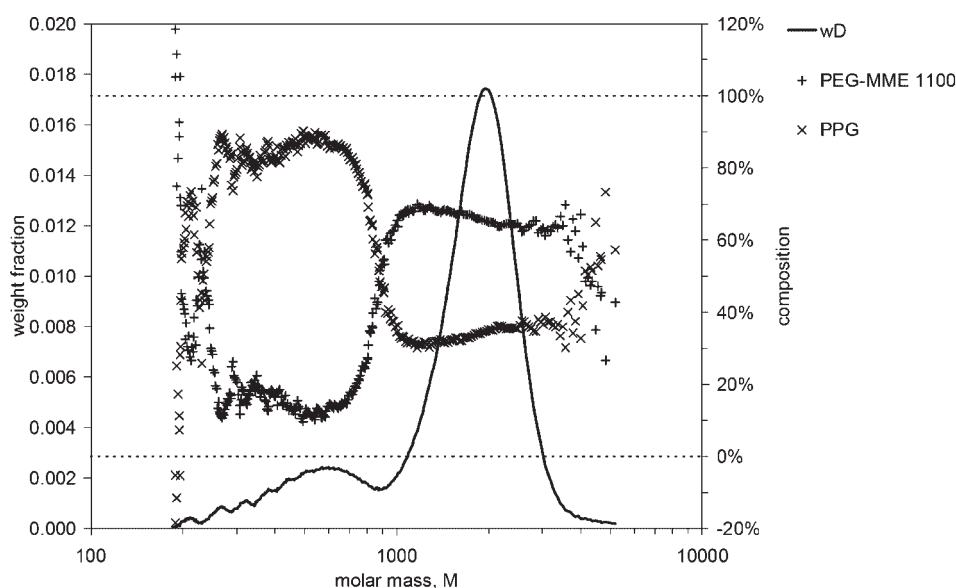


Figure 4. Molar-mass distribution (MMD) and chemical composition of a block copolymer based on PEG-MME 1 100, as determined by SEC with dual detection.

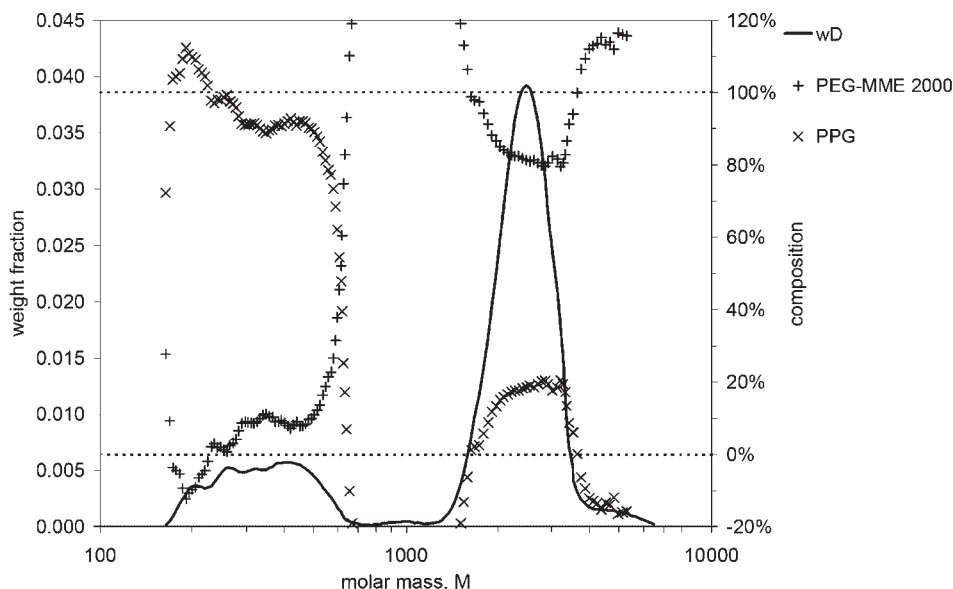


Figure 5. Molar-mass distribution (MMD) and chemical composition of a block copolymer based on PEGMME 2 000, as determined by SEC with dual detection.

which had entered the copolymer or formed a homopolymer.

From SEC, no information on the functionality of the PPG can be obtained (diol or monoallyl-ether). This can only be achieved using LCCC (at the CAP for the PO unit).

As we have shown previously,^[38,39] critical conditions for the PO unit exist in two different systems, which allow a separation of PEG in SEC mode: Discovery HS PEG in 44% acetone or Jordi RP500 in 90% acetone-water.

In the latter column, an even better separation of PEG, PPG diols and allylic PPG is achieved in THF-water (with a critical point for the PO unit at 80 wt.-% THF).

In Figure 6 the molar masses of different polymers (PEGs, PPGs and allyl terminated PPGs which had been synthesized for comparison by propoxylation of allylic alcohol) are plotted versus their elution volume (see Table 3).

This graph clearly shows that PEGs elute in SEC mode, while PPG diols and allyl-

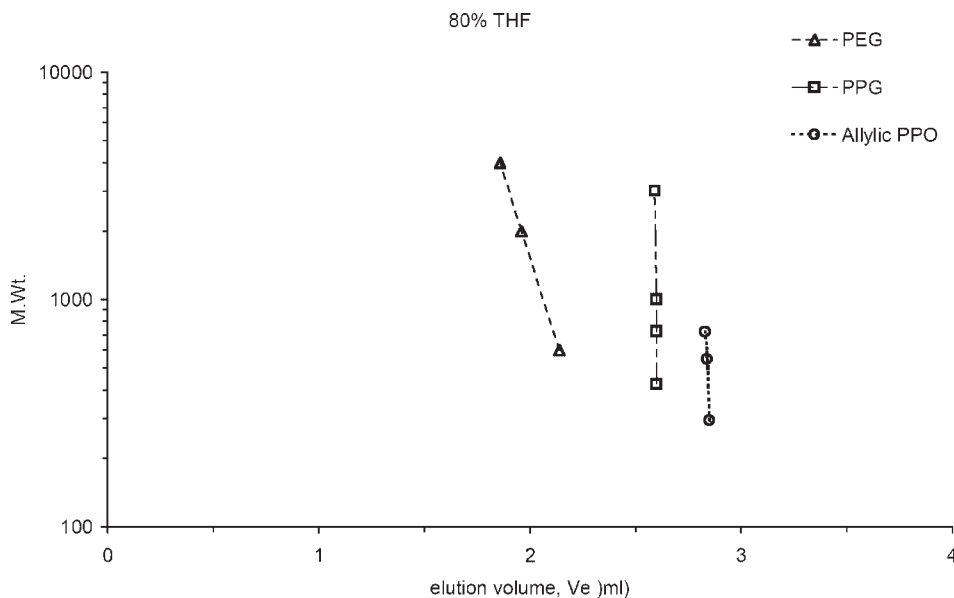


Figure 6. LCCC: molar mass versus eluent volume (V_e), as obtained on the Jordi column in 80 wt.-% THF-water, flow rate $0.5 \text{ ml} \cdot \text{min}^{-1}$, detection: ELSD.

terminated PPGs elute at the CAP (at 2.60 ml and 2.84 ml, respectively), which is sufficient to separate both homologous series (at least in the molar-mass range that is relevant in this study).

A practical example is shown in Figure 7: under these conditions, the block copolymer elutes as the starting block

oligomers, which should be present in such a sample in considerable quantities.

The allylic PPG is sufficiently separated from the solvent peak, which means, that it can be quantified from RI detection if the molar-mass dependence of response factors is known. This is, however, not easily achieved, because no sufficiently-pure reference materials are available. Moreover, the effect of preferential solvation has to be taken into account in mixed mobile phases.

Obviously, quantification in LCCC is more difficult than in SEC, which is always performed in single eluents. In SEC with coupled density and RI detection, both fractions can be quantified quite well, if they do not overlap too much. As has been shown in previous papers,^[33,37,43] the response factors of density and RI detection depend on chemical composition and molar mass.

When a mass m_i of a copolymer, which contains the weight fractions w_A and w_B (where $w_B = 1 - w_A$) of the components A and B, elutes

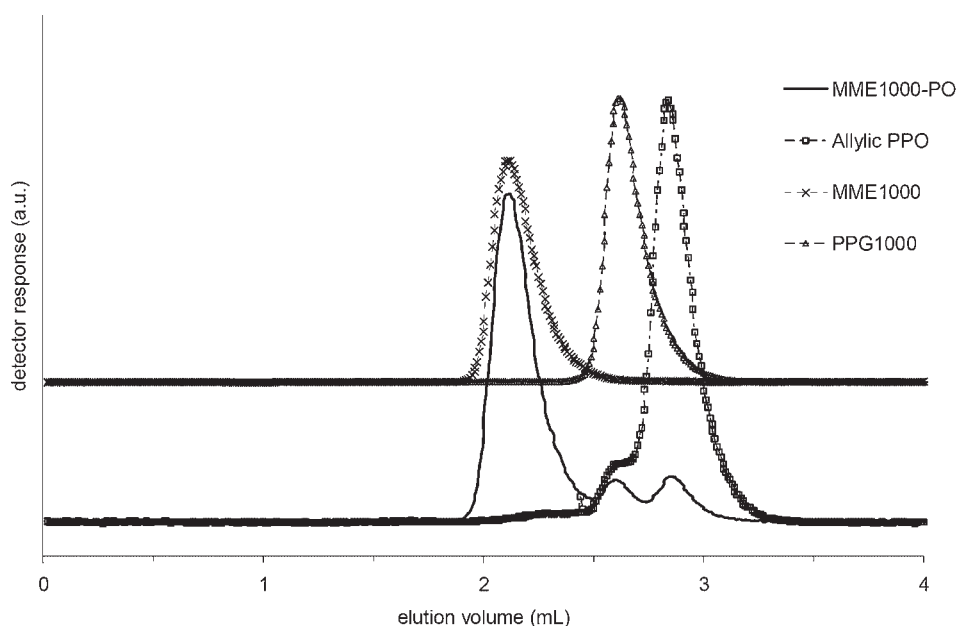


Figure 7. LCCC of a diblock copolymer (MME 1000-PO) which had been synthesized by propoxylation of PEG-MME 1000, the starting material and the possible side products, as obtained on the Jordi column in 80 wt.-% THF.

in the interval i of the peak, its composition (w_A) can be calculated from the areas $x_{i,j}$ of the slice obtained from both detectors, and the corresponding response factors $f_{j,A}$ and $f_{j,B}$, wherein j denotes the individual detectors

$$\frac{1}{w_A} = 1 - \frac{\left(\frac{x_{i,1}}{x_{i,2}} * f_{2,A} - f_{1,A}\right)}{\left(\frac{x_{i,1}}{x_{i,2}} * f_{2,B} - f_{1,B}\right)} \quad (3)$$

Once the composition of a fraction is known, its mass can be calculated using Equation (4).

$$m_i = \frac{x_{i,1}}{w_A * (f_{1,A} - f_{1,B}) + f_{1,B}} \quad (4)$$

The response factors of a polymer depend on molar mass if the end groups are chemically different from the repeat unit. This dependence can be described by

$$f_i = f_\infty + \frac{K}{M_i} \quad (5)$$

wherein f_i is the response factor of a molecule with the molar mass M_i , f_∞ is the response factor of the repeat unit (or a molecule with infinitely or at least very high molar mass), and K is a constant describing the influence of the end group.

The molar mass of the copolymer fraction is high enough to make the molar-mass dependence negligible. Hence the amount and composition of the copolymer fraction were calculated using Equation (3) and Equation (4).

The homopolymer fraction, however, can be quantified using Equation (5), if the molar mass dependence of response factors is known. K and f_∞ can be determined by linear regression, if a sufficient number of monodisperse standards with different molar masses is available.

As a good approximation, one may also use a two-point calibration (in this case using high-molecular-weight PPG and allyl alcohol), or an iteration procedure (using a high-molecular-weight polymer and a poly-disperse low-molecular-weight sample).^[35–37] In this study, we determined K by the iteration procedure.

We integrated both fractions in SEC separately and

calculated their amount and composition of the copolymer from dual detection and the amount of the homopolymer using Equation (5). Therefrom we calculated the amount of PO, which had been incorporated in the copolymer, and the amount, which had formed a homopolymer (PPG diols and allyl terminated PPG).

There are different factors which affect the quantity of unwanted homopolymers. Some show only a minor effect, but some have significant effect on the copolymer composition.

NaH Quantity

The concentration of NaH determines the concentration of active centers. Consequently, the amount of NaH present will influence the time required for complete consumption of the monomer, as the reaction rate is directly proportional to the concentration of active centers. On the other hand, the concentration of active centers (in presence of a large excess of hydroxy groups) cannot be expected to have a significant effect on the molar-mass distribution and on the formation of homopolymers. This is indeed the case, as can be seen in Figure 8 and Figure 9.

Temperature

One may speculate, that the temperature dependence of the rate constants for propagation and chain transfer may be different. In this case, the formation of allyl-terminated homopolymers could be influenced by a variation of the

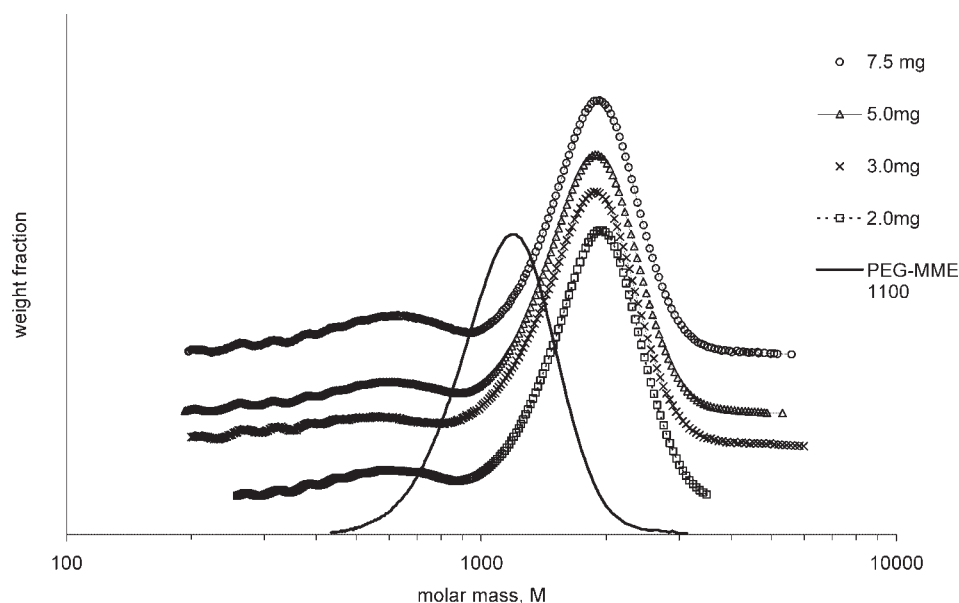


Figure 8. Effect of NaH quantity on MMD, as obtained from SEC on Phenogel 100 Å. Reaction conditions: experiments #1, 5, 6, 8 in Table 1.

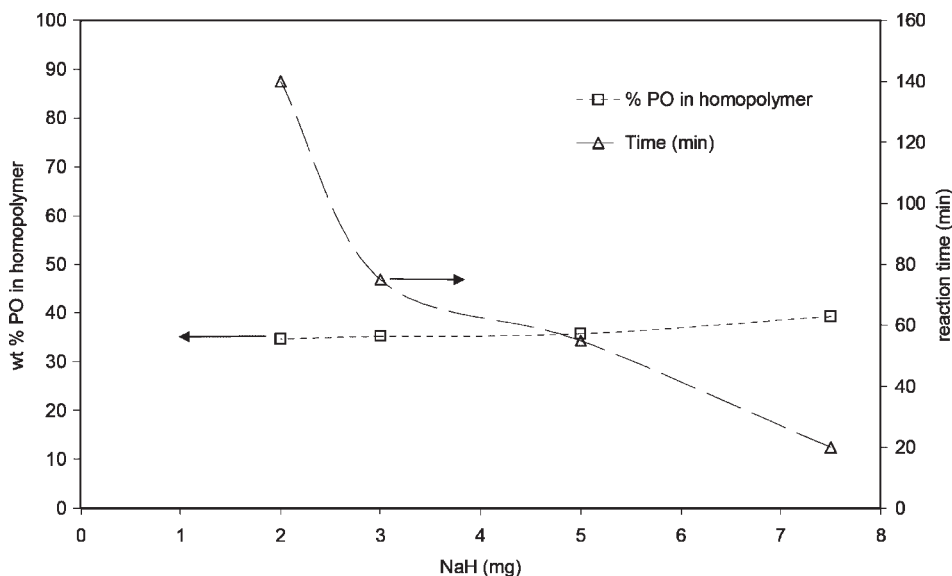


Figure 9. Effect of NaH quantity. Reaction conditions: experiments #1,5,6,8 in Table 1.

polymerization temperature. As can be seen in Figure 10 and Figure 11, the formation of unwanted homopolymers somewhat decreases with reduced temperature, but for the sake of dramatically-increased reaction times.

Concentration of Hydroxy Groups

According to the findings of Becker^[20] a high concentration of hydroxy groups can more or less suppress the chain-transfer reaction. Therefore we used PEG-MME with three different molecular weights as the starting material, corresponding to a different concentration of terminal

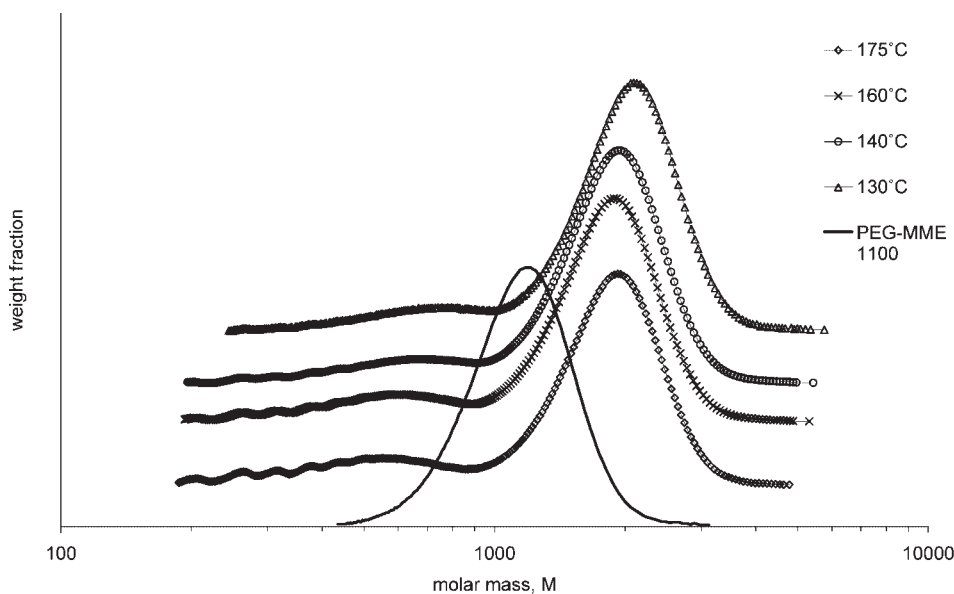


Figure 10. Effect of reaction temperature on MMD, as obtained from SEC on Phenogel 100 Å. Reaction conditions: experiments #2, 3, 4, 5 in Table 1.

hydroxy groups (see Table 1 and Table 4). All amounts of the reagents and other synthetic protocols were the same as for the sample above, except for the different number of hydroxy groups. The effect of the different -OH concentration on homopolymer formation can be seen in Table 4 and Figure 12: with increasing -OH concentration the amount of (allyl-terminated) homopolymer decreases.

Concentration of Monomer in Feed

It is obvious that a lower monomer concentration will have the same effect as a higher -OH concentration. As can be seen in Figure 13 and Figure 14, the monomer content in the feed has an enormous effect on the formation of the unwanted homopolymer. By keeping the PO concentration very low, the formation of allyl-terminated homopolymers can be suppressed. This suggests a possible solution: if the monomer is added approximately at the rate at which it is consumed, one may obtain fairly-clean block copolymers. Unfortunately our current experimental setup does not allow the continuous slow addition of monomer. However, if the polymerization can be performed under continuous-flow conditions,^[14,44] the monomer concentration can be kept low enough to probably suppress the formation of the allyl-terminated homopolymer almost completely. This will be the subject of another communication.

Polarity of the Medium

Polymerization of PO by an anionic mechanism should be strongly influenced by the polarity of the medium, which could promote or suppress dissociation. It may be assumed, that there may be a difference in the rate constants, if the reaction proceeds through ion pairs or free ions. Therefore we used several solvents

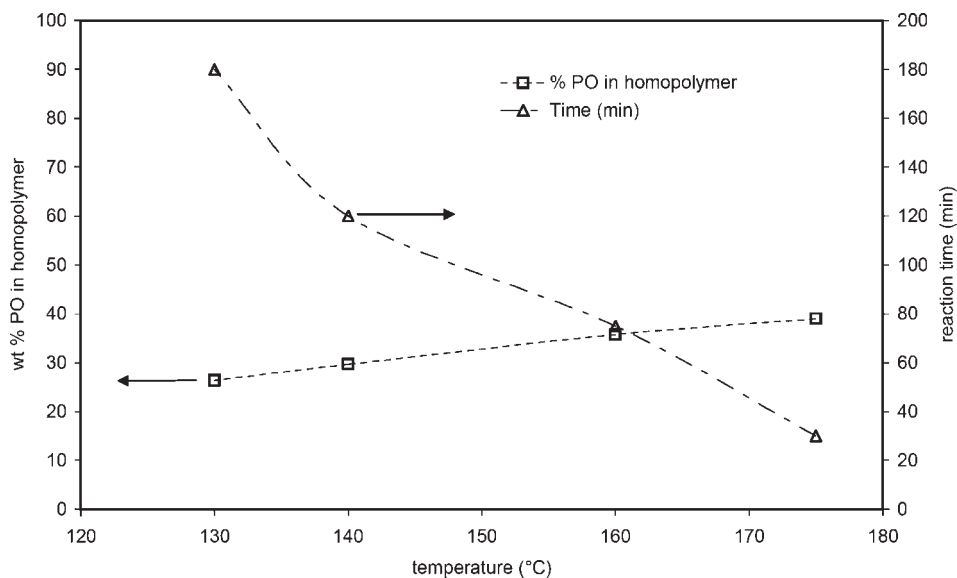


Figure 11. Effect of reaction temperature. Reaction conditions: experiments #2, 3, 4, 5 in Table 1).

Table 4. Effect of $-\text{OH}$ content ($\text{mmol} \cdot \text{g}^{-1}$) of feed.

Experiment number	Average molar mass of PEG-MME	$-\text{OH}$ content in feed	Amount of converted PO in homopolymer
	$\text{g} \cdot \text{mol}^{-1}$	$\text{mmol} \cdot \text{g}^{-1}$	%
8	1 100	0.50	29.77
9	550	0.91	8.48
10	2 000	0.25	39.96

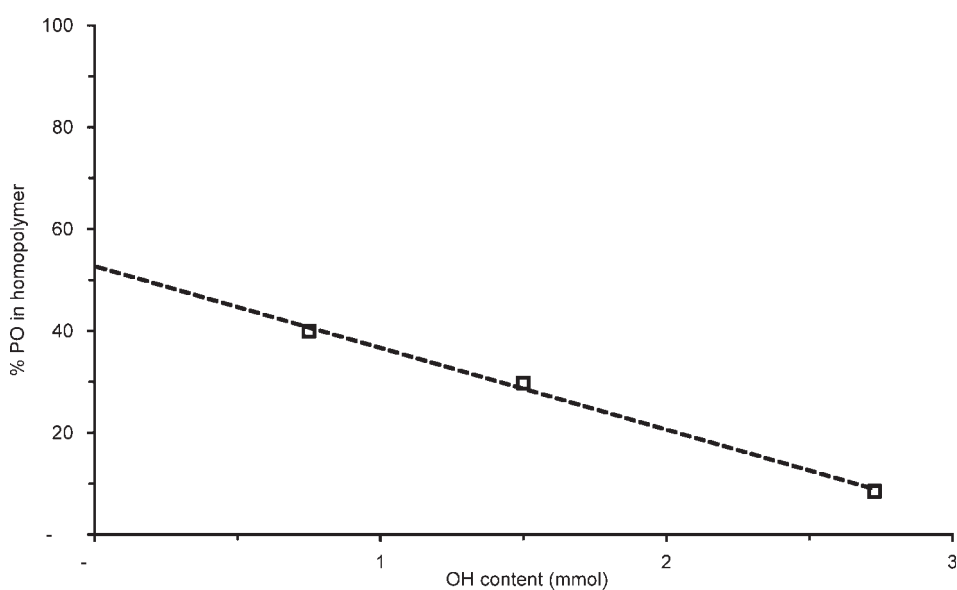
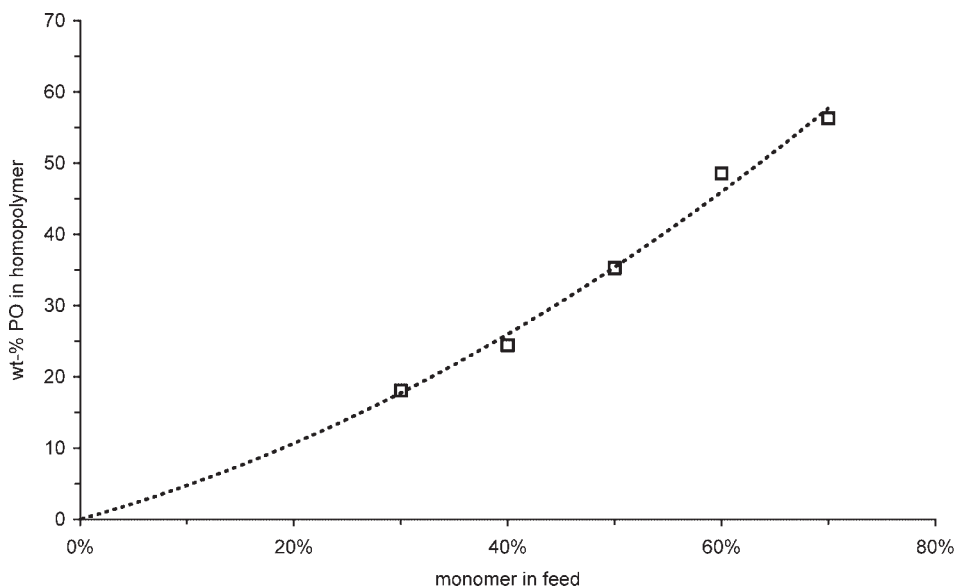


Figure 12. Influence of the $-\text{OH}$ content ($\text{mmol} \cdot \text{g}^{-1}$ of feed) on the formation of homopolymers (experiments #8, 9, 10 in Table 1 and Table 4).



■ Figure 13. Effect of monomer ratio (weight fraction) in feed (experiments #7, 8, 11, 12, 13 in Table 1).

with different polarity. Polymerizations were performed at the same temperature (160 °C) and for the same time (180 minutes).

As can be seen in Figure 15, in a highly-non-polar solvent like cyclohexane, almost no conversion (to copolymer or homopolymer) is observed after this period; in a more-polar solvent like dioxane, more PO goes to homopolymer and less to desired block copolymer. In a

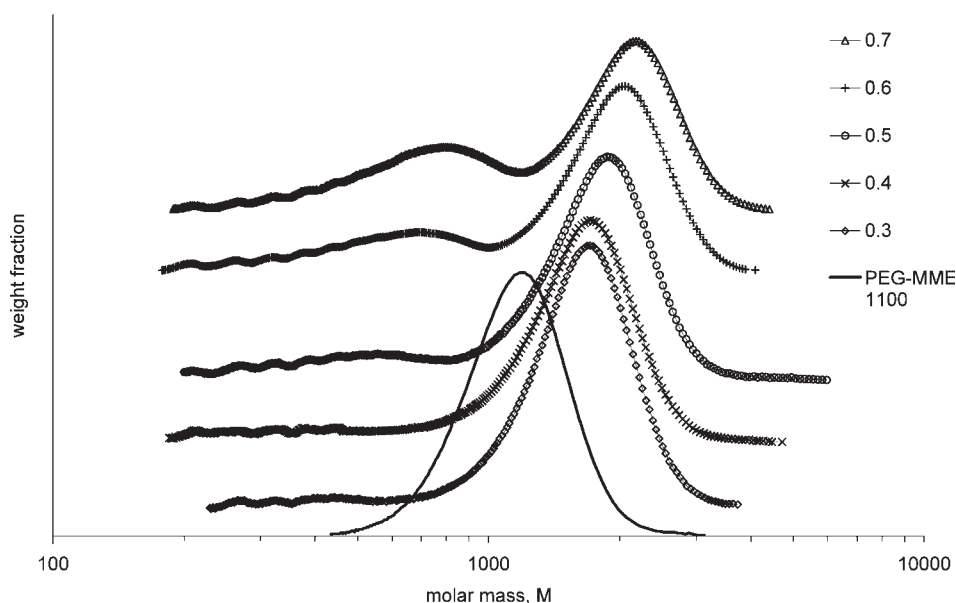
solvent of intermediate polarity like toluene the reaction proceeds more slowly, but there is somewhat less homopolymer formed. As the next step we used different concentrations of toluene in the reaction mixture and found that more toluene in the mixture results in a lower molar mass of homopolymer and copolymer (which means less conversion), as can be seen by the shift of the maxima in Figure 16.

Comparison with Conductive Heating

Microwave reactors can enhance the reaction rates (compared to conductive heat-

ing) if the reactants absorb microwave radiation. Recently it was reported that growing polymer chains strongly absorb microwaves, thus considerably higher polymerization rates are observed with microwave irradiation instead of conductive heating.^[45]

For comparison we performed an experiment keeping all conditions the same and just changing the mode of heating. The first, obvious advantage one can observe



■ Figure 14. Effect of monomer ratio (weight fraction) in feed. on MMD, as obtained from SEC on Phenogel 100 Å. Reaction conditions: experiments #7, 8, 11, 12, 13 in Table 1.

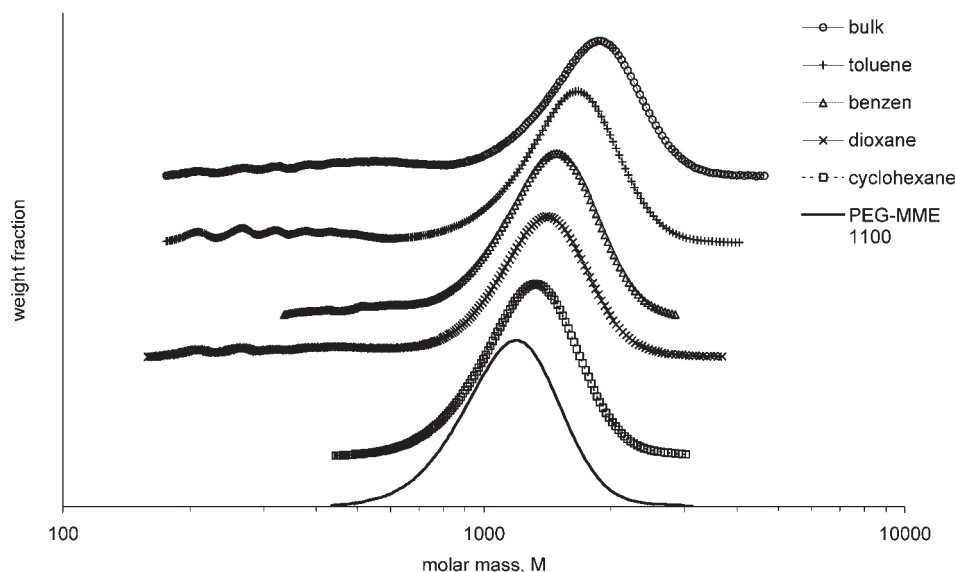


Figure 15. MMD of products polymerized in different solvents (from SEC on Phenogel 100 Å). Reaction conditions: experiments #14, 15, 16, 17 in Table 2 and #8 in Table 1).

during synthesis is a considerable reduction in the reaction time. Using microwave irradiation, completion of the reaction takes about 1 fifth of the time required with conductive heating in an autoclave.

MMD comparison by SEC of both experiments in Figure 2 shows that in the microwave experiment a considerably smaller amount of propylene oxide goes to homopolymer than with conductive heating. The reason for this may be the more homogeneous temperature in the microwave experiment.

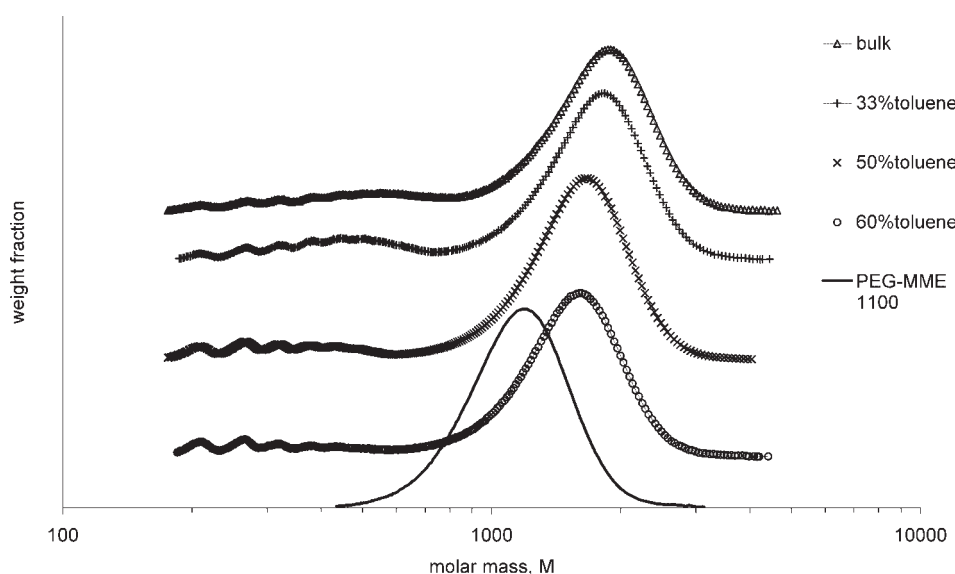


Figure 16. MMD of products synthesized at 160 °C in microwave (experiment #8) and in autoclave (PEG-MME 1000 = 30.65g, PO = 30.65g, NaH = 61.35g).

Conclusion

PEO-*block*-PPO diblock copolymers can be synthesized in bulk using controlled microwave dielectric heating in sealed vessels. The required reaction time is reduced by a factor of 5 by doing it under microwave irradiation. This time-saving aspect makes it feasible to study the effects of many different parameters on the quality of the polymer products. In the present manuscript we have investigated the effects of different parameters on product quality and found that the concentration of the monomer and the hydroxy content in the feed, the polarity of reaction medium and the

temperature can all affect the overall quality of the polymer. It can be concluded that it is possible to prepare fairly-clean block copolymers if the concentration of the monomer can be kept very low.

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